



CRANE CLASSIFICATIONS



CoreLifting.com





Bridge Crane Classifications

ASME Hoist Duty Classifications

Load Classes and Cycles

**Comparison chart for European
and CMAA Crane Classifications**

As interpreted by Core Lifting LLC.



Bridge Crane Classifications

Crane Class A: Standby Service

Designed for cranes placing machinery during the installation of the machinery, and then for maintenance of that machine. After erection of the crane, and installation of the machine it is used for servicing is complete, the crane may stay idle for long periods of time. This service rating allows for 2-5 nearly to full capacity lifts per hour. Speeds should be slow.

Typical Applications for this type of Crane:

Power plants, transformer repair centers, pump rooms and other areas where maintenance of the machinery below it is the cranes' main purpose.

Crane Class B: Light Service

No specific operator assigned to be using this crane. Capable of up to 10 lifts near or at capacity in an hour. A crane in this class may be idle for extended periods of time, however when it is used it may be in constant operation. Speeds should be slow

Typical Applications for this type of Crane:

Light assembly, service buildings, light warehouse service

Crane Class C: Moderate Service

Capable of 5-15 near to full capacity lifts per hour. Crane design life and construction should permit usage in the average of loads being 50% of rated capacity. In terms of numbers, most cranes are built to meet Class C service requirements

Typical Applications for this type of Crane:

Machine shops, paper mills, oilfield service companies

Crane Class D: Heavy Duty Service

Top riding bridge and gantry type cranes capable of lifting 10-20 nearly to full capacity lifts per hour. Speeds are medium to fast. 65% or less of the lifts at rated or near rated capacity

Typical Applications for this type of Crane:

Heavy machine shops, steel warehouses, stamping plants, lumber plants

Crane Class E: Severe Duty Service

Continuous material handling. Capable of 20-40 lifts per hour at or near capacity. Crane will be high speed and cycles for each motion need to be clearly defined. Crane designed for minimum down time.

Typical Applications for this type of Crane:

Scrap yards, magnet bucket cranes, coal handling, steel mills

Crane Class F: Continuous Severe Service (Steel Mill)

Continuous material handling. Designed for near capacity lifts at all times. Crane will be high speed and cycles for each motion need to be clearly defined. Crane designed for minimum down time.

Typical Applications for this type of Crane:

Steel mills, scrap yards, magnet bucket cranes, coal handling



ASME Hoist Duty Classifications

ASME Duty Class	Service Classification	Description
H1	Standby or Infrequent	Idle 1-6 months, used for installation and/or maintenance
H2	Light	Light maintenance, fabrication; capacity loads infrequently handled; randomly distributed loads; low running time
H3	Standard	General machining; randomly distributed loads; total running time $\leq 25\%$ of working period
H4	Heavy	High volume handling of heavy loads near rated capacity; steel warehousing, machining, foundries; total running time $\leq 50\%$ of working period
H5	Severe	Bulk material handling with attachments; approach continuous operation

Hoist Class	Bearing Life at $K=0.65$ (hours)	Uniformly Distributed Working Periods		Infrequent Heavy Works Periods	
		Max On Time (Minutes Per Hour)	Max Starts Per Hour	Minutes	Starts
H1	1250	7.5	75	15	100
H2	2,500	7.5	75	15	100
H3	5,000	15	150	30	200
H4	10,000	30	300	30	300
H5	20,000	60	600	Not Applicable	Not Applicable



Load Class and Cycles

Load Class	Load Cycles			
	N1	N2	N3	N4
L ₁	A	B	C	D
L ₂	B	C	D	E
L ₃	C	D	E	F
L ₄	D	E	F	F
	Occasional Use	Regular Use – Intermittent Operation	Regular Use Continues Operation	Regular Use, Severe Operation
Load Classes L₁=Cranes which hoist rated load – Very Light Loads L₂=Cranes that rarely hoist rated load and normal loads 1/3 of the time L₃=Cranes which hoist the rated load fairly frequently and normally – 1/3 and 2/3 of rated load L₄= Cranes which are regularly loaded with full capacity Load Cycles N₁=20,000-100,000 cycles N₂=1000,000-500,000 cycles N₃=500,000 to 2,000,000 cycles N₄=Over2,000,000 cycles				



CMAA Crane Classifications and FEM Comparison Chart

Crane Classification	Hoist Classification			Use	Typical Operation	Load Spectrum	Lifts per Hour	Applications
CMAA	FEM	HMI	ISO					
Class A	1C _m	H1	M2	Light Duty	One Shift	Light Duty	2 Per Hour	Standby - infrequent Service
Class B	1B _m	H2	M3	Light Duty	One Shift	Max Load Rarely Lifted	2 Per Hour	Small repair shops, light duty fabrication
Class C	1A _m	H3	M4	Light to Medium Duty	One Shift	Max Load Occasionally	2-5 lifts per hour	Repair shops, fabrication service buildings warehousing
Class D	2 _m	H4	M5	Medium to Heavy Duty	Up to two shifts	Medium to Heavy Loads	5-10 Lifts per Hour	Heavy machine shops, steel warehousing, lumber mills,
	3 _m	H4	M6	Heavy Duty	Two Shifts	Heavy Loads	10-20 Lifts per Hour	Heavy machine shops, steel warehousing, lumber mills,
Class E	4 _m	H4/H5	M7	Extremely Heavy Duty	Two to three shift	Regular Heavy Loads	Over 20 lifts an hour	Magnet, scrap yard, container handling, steel mill
Class F	6	N/A	M8	Extreme Heavy Duty with Continual Operation				